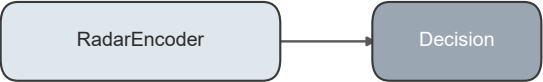


(a) Four architectures compared

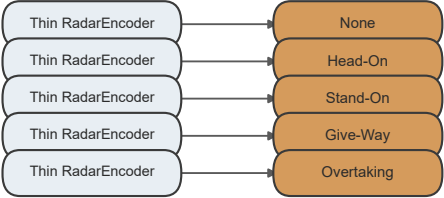
Single network

369K · one perception, one decision



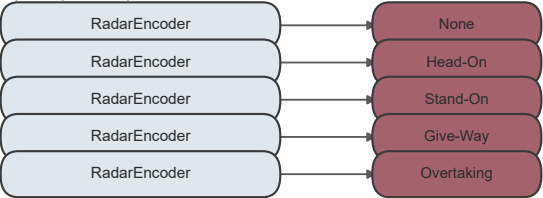
Separate experts, thin

363K · perception duplicated x5, each narrowed



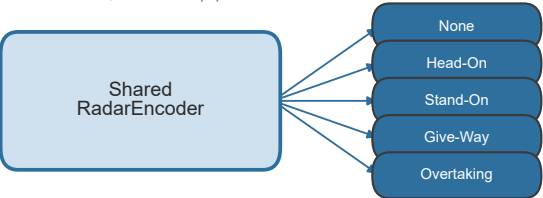
Separate experts, full

1.83M · perception duplicated x5, full width



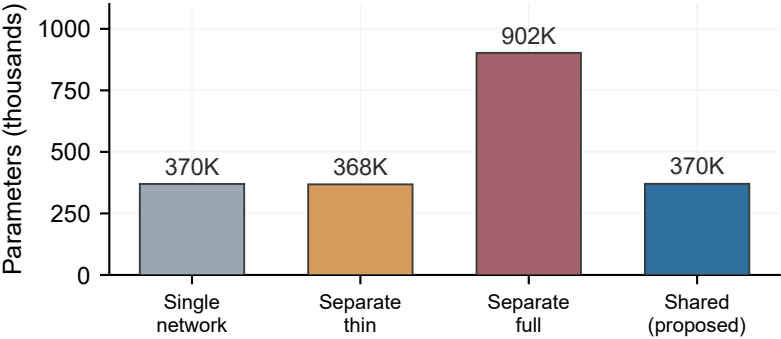
Shared perception (proposed)

370K · shared trunk, residual $\Delta\mu$ per situation

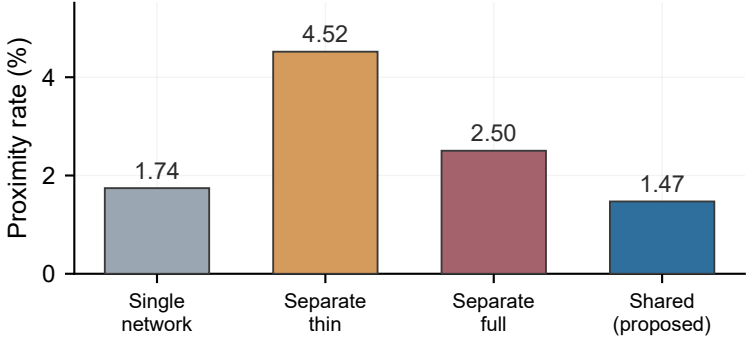


Radar perception is ~89% of single-network parameters.
The same expert decomposition is applied to MessageActor, ControlActor and Critic.

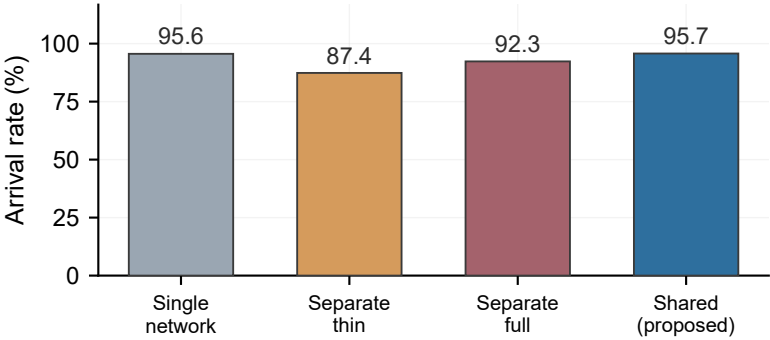
(b) Parameter count



(c) Proximity rate



(d) Arrival rate



v12mix: shared MoE (shared eye/backbone, situation steering heads, train soft route-mix=0.15) vs single / separate-thin / separate-full, 3 seeds.
SHARED ≈ SINGLE in trainable params (eye shared); THICK duplicates encoders. SHARED ≥ SINGLE on arrival/proximity; THIN/THICK lose. Not residual $\Delta\mu$.